CoreTech Cybersecurity Internship Task

Introduction

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Networking Fundamentals for Cybersecurity

Introduction

Networking is the foundation of cybersecurity. Understanding how devices communicate, how data travels across networks, and how attackers exploit network weaknesses is essential for protecting systems and information.

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1. OSI Model (Open Systems Interconnection Model)

The OSI Model consists of **7 layers** that describe how data travels from one device to another.

```
+---------------------+
| 7. Application      |
+---------------------+
| 6. Presentation     |
+---------------------+
| 5. Session          |
+---------------------+
| 4. Transport        |
+---------------------+
| 3. Network          |
+---------------------+
| 2. Data Link        |
+---------------------+
| 1. Physical         |
+---------------------+
```

Layer 1 – Physical Layer

- Transmits raw bits through cables, radio signals, or fiber optics.
- Deals with hardware components.

Examples: Ethernet cable, Hub, Fiber optic cable.

Layer 2 – Data Link Layer

- Transfers data between devices on the same network.
- Uses MAC addresses.

Examples: Switches, Ethernet.

Layer 3 – Network Layer

- Responsible for routing data between networks.
- Uses IP addresses.

Examples: Routers, IP Protocol.

Layer 4 – Transport Layer

- Ensures reliable or fast data delivery.
- Uses TCP and UDP.

Examples: TCP, UDP.

Layer 5 – Session Layer

- Establishes, manages, and terminates communication sessions.

Example: Login sessions.

Layer 6 – Presentation Layer

- Translates data formats.
- Handles encryption and compression.

Examples: SSL/TLS, JPEG, ASCII.

Layer 7 – Application Layer

- Provides network services directly to users.

Examples: HTTP, HTTPS, FTP, DNS.

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2. TCP/IP Model

The TCP/IP model is the practical networking model used on the Internet.

```
+---------------------+
| Application Layer   |
+---------------------+
| Transport Layer     |
+---------------------+
| Internet Layer      |
+---------------------+
| Network Access Layer|
+---------------------+
```

Application Layer

Provides services to applications.

Examples:

- HTTP
- HTTPS
- FTP
- DNS
- SSH

Transport Layer

Responsible for end-to-end communication.

Protocols:

- TCP
- UDP

Internet Layer

Handles routing and IP addressing.

Protocols:

- IP
- ICMP

Network Access Layer

Handles physical transmission of data.

Examples:

- Ethernet
- Wi-Fi

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3. Common Network Protocols

HTTP (HyperText Transfer Protocol)

- Used for website communication.
- Data is transmitted in plain text.
- Default Port: **80**

Example:

```
http://example.com
```

HTTPS (HyperText Transfer Protocol Secure)

- Secure version of HTTP.
- Uses SSL/TLS encryption.
- Default Port: **443**

Example:

```
https://example.com
```

FTP (File Transfer Protocol)

- Used for file transfer.
- Not encrypted by default.
- Default Port: **21**

SSH (Secure Shell)

- Secure remote system access.
- Encrypts communication.
- Default Port: **22**

DNS (Domain Name System)

- Converts domain names into IP addresses.

Example:

```
google.com
      ↓
142.250.x.x
```

DHCP (Dynamic Host Configuration Protocol)

- Automatically assigns IP addresses to devices.

Provides:

- IP Address
- Subnet Mask
- Gateway
- DNS Server

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4. IP Addressing and Subnetting Basics

What is an IP Address?

An IP address uniquely identifies a device on a network.

Example:

```
192.168.1.10
```

IPv4 Structure

```
192.168.1.10
```

Contains four octets.

Range:

```
0 - 255
```

Public vs Private IP

Public IP

Used on the Internet.

Example:

```
8.8.8.8
```

Private IP

Used inside local networks.

Ranges:

```
10.0.0.0 - 10.255.255.255
```

```
172.16.0.0 - 172.31.255.255
```

```
192.168.0.0 - 192.168.255.255
```

Subnetting Basics

Subnetting divides a network into smaller networks.

Example:

Network:

```
192.168.1.0/24
```

Subnet Mask:

```
255.255.255.0
```

Meaning:

- Network Address: 192.168.1.0
- Host Range: 192.168.1.1 – 192.168.1.254
- Broadcast Address: 192.168.1.255

5. Difference Between TCP and UDP

Feature	TCP	UDP
Connection	Connection-Oriented	Connectionless
Reliability	Reliable	Less Reliable
Error Checking	Yes	Basic
Speed	Slower	Faster
Acknowledgment	Required	Not Required
Usage	Web, Email, File Transfer	Streaming, Gaming, VoIP

TCP Process

```
Client ---- SYN ----> Server
Client <--- SYN-ACK -- Server
Client ---- ACK ----> Server
```

Called the **Three-Way Handshake**.

6. Types of Network Attacks

Man-in-the-Middle (MITM)

An attacker secretly intercepts communication between two parties.

```
User <----> Attacker <----> Server
```

Risks:

- Data theft
- Credential theft
- Session hijacking

DNS Spoofing

The attacker provides a fake DNS response.

```
User → google.com
       ↓
Fake IP Address
```

```
        ↓
Malicious Website
```

Risks:

- Phishing attacks
- Malware distribution

ARP Poisoning

The attacker sends fake ARP messages to associate their MAC address with another device's IP.

```
Victim → Attacker → Router
```

Risks:

- Traffic interception
- Session hijacking
- Data theft

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7. How Firewalls Work

A firewall monitors and controls incoming and outgoing traffic according to security rules.

Firewall Diagram

```
Internet
    |
    |
[ Firewall ]
    |
    |
Internal Network
```

Functions

- Block unauthorized access
- Filter traffic
- Monitor network activity
- Enforce security policies

Types

1. Packet Filtering Firewall

2. Stateful Firewall
3. Next-Generation Firewall (NGFW)

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8. How Routers Work

A router connects multiple networks and forwards data packets between them.

Router Diagram

```
        Internet
            |
        [ Router ]
        /        \
       /          \
 PC 1              PC 2
```

Functions

- Route traffic between networks
- Assign local IP addresses
- Perform NAT (Network Address Translation)
- Connect LAN to the Internet

Packet Routing Example

```
PC → Router → ISP → Internet → Website
```

Conclusion

Networking fundamentals are essential for cybersecurity professionals. Understanding the OSI and TCP/IP models, network protocols, IP addressing, subnetting, TCP vs UDP, common network attacks, and the role of firewalls and routers helps security analysts identify vulnerabilities, secure communications, and defend networks against cyber threats